

CS/BC/AI/IT-401
Examination –May- 2024

B.Tech. IV Sem : CS/BC/AI/IT
Computer Network

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) i. A network that is simulated using software and does not physically exist is known as a:

A) Wired Network B) Wireless Network C) Hybrid Network D) Virtual Network

ii. Which of the following is a characteristic of a PAN (Personal Area Network)?

A) Covers a large geographical area B) Typically owned and operated by multiple organizations C) Connects devices within a home or office D) Uses satellite communication

(b) The OSI model provides a _____ framework for network communication. The _____ layer is responsible for establishing, maintaining, and terminating connections between nodes.

(c) Define connection-oriented services in computer networks and provide examples of protocols that use this approach.

OR

Discuss the importance of the OSI model in network troubleshooting in short.

(d) Discuss the OSI model and its significance in understanding how data is transmitted across networks?

OR

Discuss the key components of a computer network architecture and their respective roles in facilitating communication.

Marks	BL	CO
02	1,1	1,1
02	1	2
03	1	3

03	2	3
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05	2	3
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05	2	1
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Q.2 (a) i. Which network device operates at the Data Link layer of the OSI model and filters traffic based on MAC addresses?

A) Repeater B) Bridge C) Switch D) Gateway

ii. Which type of hub regenerates and retransmits signals?

A) Active hub B) Passive hub C) Repeater hub D) Router hub

02	1,1	3,2
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- (b) A bridge is used to connect two _____ together. Twisted pair cables are an example of _____ transmission media.
- (c) Discuss the three different types of transmission media with their bandwidth capabilities briefly.

02

1	2
1	2

03

OR

Discuss the importance of using standardized connectors in network design in short.

03

1	2
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- (d) Differentiate between active and passive hubs, including their functions and when each might be used in a network.

05

2	2
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OR

Explain the concept of a mesh network topology and describe when it might be used in a network design.

05

2	1
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Q.3 (a) i. Which of the following best describes the purpose of pipelining?

02

- A) To reduce the number of transmission errors
 B) To increase the speed of data processing by overlapping processing stages
 C) To improve the security of data transmission
 D) To simplify the network protocol

1,1	1,1
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ii. Which of the following statements about piggybacking is true?

- A) It increases the number of acknowledgments in a network.
 B) It reduces the efficiency of data transmission.
 C) It is used primarily in simplex communication.
 D) It improves the efficiency of data transmission by utilizing otherwise idle transmission opportunities.

- (b) ALOHA was later improved to _____ ALOHA to increase efficiency. CSMA/CD is used in Ethernet networks to detect _____ and take corrective actions to avoid them.

02

1	3
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- (c) What is the primary function of the Data Link Layer in the OSI model?

03

2	3
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OR

Explain the difference between static and dynamic channel allocation in the context of network communication.

03

2	1
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- (d) Describe the stop-and-wait flow control technique. Explain how it ensures reliable communication between a sender and a receiver, and discuss its advantages and limitations compared to other flow control techniques.

05

2	3
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OR

A radio station is using 14.44 kbps channel for message transmission and the sending message packets are 100 bits long. Calculate the maximum throughput for the channel using slotted aloha.

05

5	4
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Q.4 (a) i. What does the Bellman-Ford algorithm return if a negative cycle is detected in the graph?

02

- A) The shortest path from the source node to all other nodes.
- B) The presence of a negative cycle in the graph.
- C) Infinity for all nodes reachable from the negative cycle.
- D) An error indicating that the graph contains a negative cycle.

ii. What is the purpose of fragmentation in the network layer?

- A) To detect and correct errors in data packets
- B) To reassemble data packets at the destination
- C) To divide large data packets into smaller fragments for transmission
- D) To manage congestion in the network

(b) i. Which of the following is a key objective of routing algorithms?

02

- A) Minimizing network latency
- B) Maximizing network bandwidth
- C) Minimizing network cost
- D) Maximizing network security

ii. Which of the following is a distance-vector routing algorithm?

- A) OSPF (Open Shortest Path First)
- B) BGP (Border Gateway Protocol)
- C) RIP (Routing Information Protocol)
- D) EIGRP (Enhanced Interior Gateway Routing Protocol)

(c) Explain how congestion control mechanisms help in preventing network congestion and ensuring efficient data transmission in short.

03

OR

How does the network layer facilitate communication between devices on different networks?

03

(d) Explain Dijkstra's algorithm for finding the shortest path in a graph with the help of an example.

05

OR

Compare and contrast IPv4 and IPv6 in terms of addressing, address space, header format, and adoption. Describe the advantages of IPv6 over IPv4 and the challenges in migrating from IPv4 to IPv6.

05

Q.5 (a) i. Which of the following statements about UDP (User Datagram Protocol) is true?

02

- A) UDP provides reliable, ordered, and error-checked delivery of data packets.
- B) UDP is connection-oriented, ensuring that data packets are delivered in the correct order.

2,2	4,3
1,1	4,4
2	3

2	3
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3	4
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2	3
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1,1	3,3
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C) UDP is faster than TCP (Transmission Control Protocol) due to its minimal overhead.

D) UDP is commonly used for applications that require reliable and ordered delivery of data.

ii. What is a key feature of TCP?

A) Connectionless communication, where no connection setup is required before sending data

B) Minimal overhead to ensure high-speed data transmission

C) Flow control to manage the rate of data transmission and prevent congestion

D) Lack of support for error detection and retransmission of lost packets

(b) i. Which protocol is commonly used for accessing websites on the World Wide Web? 02

A) HTTP (Hypertext Transfer Protocol)

B) FTP (File Transfer Protocol)

C) SMTP (Simple Mail Transfer Protocol)

D) TCP (Transmission Control Protocol)

ii. SMTP is not used to deliver messages to _____

a) user's terminal

b) user's mailbox

c) user's word processor

d) user's email client

(c) Explain briefly how HTTPS provides a more secure communication as compared to HTTP? 03

OR

How does DNS resolve domain names to IP addresses?

03

2

3

(d) How does SMTP interact with local mail servers and TCP for sending and receiving emails? 05

2

3

OR

Discuss the connection establishment process in TCP and UDP. How does TCP ensure reliable data transmission compared to UDP? 05

2

3

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating
